Let's go through the answers step-by-step.

1. (a) Using standard notions, state and describe the statistical model associated with single factor ANOVA model, i.e., the effect model.

Single Factor ANOVA Model:

The single factor ANOVA (Analysis of Variance) model, also known as the one-way ANOVA model, is used to determine if there are any statistically significant differences between the means of three or more independent groups. The model can be described as:

\[Y_{ij} = \mu + \tau_i + \epsilon_{ij} \]

Where:

- \(Y_{ij} \) is the observation from the \(i\)-th treatment (or group) and the \(j\)-th replicate.

- \(\mu \) is the overall mean.

- \(\tau_i \) is the effect of the \(i\)-th treatment.

- \(\epsilon_{ij} \) is the random error associated with the \(ij\)-th observation, assumed to be normally distributed with mean zero and constant variance \(\sigma^2 \).

1. (b) Electrical conductivity data analysis

(i) Essential Assumptions:

1. **Independence**: The samples are independent of each other.

2. **Normality**: The data in each group are normally distributed.

3. **Homogeneity of Variances**: The variances among the groups are equal.

(ii) Testing if electrical conductivity is the same across all tanks

To determine if there are significant differences in electrical conductivity among the tanks at a 5% level of significance, we perform a one-way ANOVA test.

Step-by-Step ANOVA Calculation:

1. **Calculate the group means:**

- \(\bar{X}_{\text{Kala Wewa}} = \frac{265 + 262 + 275 + 269}{4} = 267.75 \)

- \(\bar{X}_{\text{Minneriya}} = \frac{278 + 274 + 256 + 265}{4} = 268.25 \)

- \(\bar{X}_{\text{Basawakkulama}} = \frac{251 + 254 + 248 + 241}{4} = 248.5 \)

- \(\bar{X}_{\text{Victoria}} = \frac{244 + 241 + 248 + 244}{4} = 244.25 \)

2. **Calculate the overall mean:**

\[\bar{X} = \frac{267.75 + 268.25 + 248.5 + 244.25}{4} = 257.6875 \]

3. **Calculate SST (Total Sum of Squares):**

\[SST = \sum_{i=1}^{4} \sum_{j=1}^{4} (Y_{ij} - \bar{X})^2 \]

Breaking down each component:

\[

\begin{align*}

SST_{Kala Wewa} &= (265-257.6875)^2 + (262-257.6875)^2 + (275-257.6875)^2 + (269-257.6875)^2 \\

SST_{Minneriya} &= (278-257.6875)^2 + (274-257.6875)^2 + (256-257.6875)^2 + (265-257.6875)^2 \\

SST_{Basawakkulama} &= (251-257.6875)^2 + (254-257.6875)^2 + (248-257.6875)^2 + (241-257.6875)^2 \\

SST_{Victoria} &= (244-257.6875)^2 + (241-257.6875)^2 + (248-257.6875)^2 + (244-257.6875)^2 \\

\end{align*}

\]

After calculation:

\[

SST = 304.6875 + 450.6875 + 181.6875 + 195.6875 = 1132.75

\]

4. **Calculate SSB (Sum of Squares Between Groups):**

\[SSB = \sum_{i=1}^{4} n (\bar{X}_i - \bar{X})^2 \]

Where \(n = 4 \) (number of observations per group):

\[

\begin{align*}

SSB_{Kala Wewa} &= 4(267.75-257.6875)^2 \\

SSB_{Minneriya} &= 4(268.25-257.6875)^2 \\

SSB_{Basawakkulama} &= 4(248.5-257.6875)^2 \\

SSB_{Victoria} &= 4(244.25-257.6875)^2 \\

\end{align*}

\]

After calculation:

\[

SSB = 408.375 + 425.875 + 84.1875 + 179.5625 = 1098

\]

5. **Calculate SSE (Sum of Squares Within Groups):**

\[SSE = SST - SSB = 1132.75 - 1098 = 34.75 \]

6. **Calculate the Mean Squares:**

$$ MSTrt = \frac{SSB}{k-1} = \frac{1098}{3} = 366 $$

$$ MSE = \frac{SSE}{N-k} = \frac{34.75}{12} = 2.8958 $$

7. **Calculate the F-statistic:**

$$ F = \frac{MSTrt}{MSE} = \frac{366}{2.8958} \approx 126.38 $$

8. **Compare the calculated F-statistic with the critical value from F-distribution table** (for $\alpha = 0.05$), $ df1 = k-1 = 3$), $ df2 = N-k = 12$)):

- Critical $ F \approx 3.49 $

Since $ F_{calculated} (126.38) > F_{critical} (3.49) $, we reject the null hypothesis.

Conclusion:

There is sufficient evidence to conclude that there is a significant difference in the electrical conductivity between the tanks at the 5% level of significance.

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2. (a) Show that in usual notations, the corrected sum of squares, SST can be written as SST = SSTrt + SSE, where SSTrt is called the sum of squares due to treatments and SSE is called the sum of squares due to error.

Using standard ANOVA notations:

- $ SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y})^2 $
- $ SSTrt = \sum_{i=1}^{k} n_i (\bar{Y}_i - \bar{Y})^2 $
- $ SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2 $

By definition,

$$ SST = SSTrt + SSE $$

This can be shown through the decomposition of total variability into variability between groups (treatments) and variability within groups (error).

2. (b) Mean squares analysis

(i) Show that \(E(MSE) = \sigma^2 \)

From ANOVA:

\[MSE = \frac{SSE}{N-k} \]

\[E(MSE) = E \left(\frac{SSE}{N-k} \right) = \frac{E(SSE)}{N-k} \]

Since \(SSE \) is the sum of squared errors:

\[E(SSE) = (N-k)\sigma^2 \]

Thus,

\[E(MSE) = \frac{(N-k)\sigma^2}{N-k} = \sigma^2 \]

(ii) Show that \(E(MSTrt) = \sigma^2 + \frac{n \sum_{i=1}^{a} \tau_i^2}{a-1} \)

From ANOVA:

\[MSTrt = \frac{SSTrt}{a-1} \]

\[E(MSTrt) = E \left(\frac{SSTrt}{a-1} \right) = \frac{E(SSTrt)}{a-1} \]

\[SSTrt = \sum_{i=1}^{a} n (\bar{Y}_i - \bar{Y})^2 \]

\[E(SSTrt) = (a-1)\sigma^2 + n \sum_{i=1}^{a} \tau_i^2 \]

Thus,

\[E(MSTrt) = \sigma^2 + \frac{n \sum_{i=1}^{a} \tau_i^2}{a-1} \]

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3. (a) Write down an appropriate statistical model using usual notations and estimate the overall mean and the treatment effects.

Statistical Model:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

Where:

- Y_{ij} is the yield from the i-th variety and the j-th plot.
- μ is the overall mean yield.
- τ_i is the effect of the i-th variety.
- ϵ_{ij} is the random error term.

Overall Mean and Treatment Effects:

Overall mean (μ):

$$\bar{Y} = \frac{\sum_{i=1}^{3} \sum_{j=1}^{3} Y_{ij}}{9} = \frac{(2+4+6) + (8+7+9) + (11+10+12)}{9} = \frac{69}{9} = 7.67$$

Treatment effects (τ_i):

$$\tau_1 = \bar{Y}_1 - \bar{Y} = \frac{(2+4+6)}{3} - 7.67 = 4 - 7.67 = -3.67$$

$$\tau_2 = \bar{Y}_2 - \bar{Y} = \frac{(8+7+9)}{3} - 7.67 = 8 - 7.67 = 0.33$$

$$\tau_3 = \bar{Y}_3 - \bar{Y} = \frac{(11+10+12)}{3} - 7.67 = 11 - 7.67 = 3.33$$

3. (b) Compute the ANOVA table and give your conclusion on the yield of different chickpea varieties at 5% level of significance.

ANOVA Table Calculation:

1. **Calculate SST (Total Sum of Squares):**

$$ SST = \sum_{i=1}^{3} \sum_{j=1}^{3} (Y_{ij} - \bar{Y})^2 $$

2. **Calculate SSB (Sum of Squares Between Groups):**

$$ SSB = \sum_{i=1}^{3} n (\bar{Y}_i - \bar{Y})^2 $$

3. **Calculate SSE (Sum of Squares Within Groups):**

$$ SSE = SST - SSB $$

4. **Calculate Mean Squares:**

$$ MSTrt = \frac{SSB}{k-1} $$

$$ MSE = \frac{SSE}{N-k} $$

5. **Calculate the F-statistic:**

$$ F = \frac{MSTrt}{MSE} $$

Conclusion:

Compare the calculated F-statistic with the critical value from the F-distribution table. If $F_{calculated} > F_{critical}$, reject the null hypothesis. This would indicate significant differences in yield among the chickpea varieties at the 5% level of significance.

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4. (a) Comments on the tensile strength using the box plot

Analyze the box plot for the hardwood concentration data to comment on the central tendency, spread, and any potential outliers in the tensile strength measurements for different hardwood concentration levels.

4. (b) Compute the ANOVA table and give conclusions at 5% significance level

Follow similar steps as in previous ANOVA calculations to fill out the ANOVA table for the tensile strength data and draw conclusions based on the F-statistic and critical values.